Construction of loss function

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Derivation the loss function of the GP mixed of continuous and categorical inputs. The general model has the input as $w = (x^t, z^t)^t$ where $x^t = (x_1, ..., x_I)$ (I continuouse variables) and $z^t = (z_1, ..., z_J)$ (J categorical variables, each variable z_i has L_i levels). Then the response $y(w)$ corresponding to the input w is

$$y(w) = \beta^t f(w) + \epsilon(w)$$

where $f(w) = (f_1(w), ..., f_m(w))^t$ is a pre-specified function in $\mathbb{R}^m$ and $\beta \in \mathbb{R}^m$ to be specified. The residual $\epsilon(w)$ is a GP process with mean zero and admit a correlation function.

Correlation function: Let w_1, w_2 be 2 input variables with the same distribution of w. We seek to compute $\text{cor}(\epsilon(w_1), \epsilon(w_2))$. Note that w_1 and w_2 consist of 2 separate components in its structure, the most common way to deal with this problem is to decompose this covariance into 2 distinc terms, that is,

$$\text{cor}(\epsilon(w_1), (\epsilon(w_2))) = \text{cor}(\epsilon(x_1, z_1), (\epsilon(x_2, z_2))) = k_{cont}(x_1, x_2)k_{cat}(z_1, z_2)$$

We define the correlation function as $k : W \times W \to \mathbb{R}$ as

$$k(w_1, w_2) = k_{cont}(x_1, x_2).k_{cat}(z_1, z_2)$$

We supposed that x has I factors and z has J factors, so we continue decomposing the above correlation function:

$$k(w_1, w_2) = \prod_{i=1}^{I} k_{cont}(x_{1i}, x_{2i}) \prod_{j=1}^{J} k_{cat}(z_{1j}, z_{2j})$$

From now, to simplify the write-up, we assume that there is only one factor for the categorical part with L levels, $\{1, 2, ..., L\}$. Okay, let K be the covariance matrix of the categorical part, with dimension $L \times L$, the (i, j) entries is equal to $k_{cat}(i, j)$.

Objective: We suppose that the kernel function of the continuous part is already known and our objective is to find an estimator for the covariance matrix in the categorical part.

Maximum likelihood: Let $W = (w_1, ..., w_n)$ be the observed variables and $Y = (y_1, ..., y_n)$ be the observed outputs. Now, we rewrite our model as

$$Y = F(W)\beta + \epsilon(W)$$

where $\epsilon(W) = (\epsilon(w_1), ..., \epsilon(w_n))^t$ and $F(W)$ is a matrix of size $n \times m$, the row i of $F(W)$ corresponds to $f(W_i)^t$. Then the likelihood or the probability distribution function of the observation Y is the n-dimensional normal distribution of mean $F(W)\beta$ and covariance matrix is similar to the one of $\epsilon(W)$. Let $T \in \mathbb{R}^{n\times n}$ be the covariance matrix of $\epsilon(W)$. The entry (i, j) of T is

$$T_{ij} = k(w_i, w_j) = \prod_{k=1}^{I} k_{cont}(x_{ik}, x_{jk}) k_{cat}(z_i, z_j)$$

Therefore, the expression for the likelihood is

$$L(\beta, T) = \frac{1}{(2\pi)^{n/2} \det(T)^{1/2}} \exp\left(-\frac{(Y - F\beta)^t T^{-1} (Y - F\beta)}{2}\right)$$

Take the logarithm of the above expression, then we get

$$l(\beta, T) = -\frac{1}{2}\left(n \log(2\pi) + \log(\det T) + (Y - F\beta)^t T^{-1} (Y - F\beta)\right)$$

Then finding maximum likelihood yields a good estimations for our hyperparameters, that is equivalent find

$$\hat{\beta}, \hat{K} = \arg \min_{\beta \in \mathbb{R}^m, K \in \mathrm{Sym}^+(L)} \log(\det T) + (Y - F\beta)^t T^{-1} (Y - F\beta)$$

To be able to do Riemannian optimization for K, this is important to rewrite T as a function of K. Note that each entry of T is written under form $T_{ij} = K_{cont}(x_i, x_j) K_{cat}(z_i, z_j)$. Firstly, we have that

$$T = K_X \circ K_Z$$

where $\circ$ is the elementwise product, $K_Z = (k_{cat}(z_i, z_j))_{1\leq i,j\leq n} \in \mathbb{R}^{n\times n}$, $K_X = (K_{cont}(x_i, x_j))_{i\leq i,j\leq n} \in \mathbb{R}^{n\times n}$. We could rewrite the matrix K_{cat} as following $K_Z = HKH^T$, where $H \in \{0, 1\}^{n\times L}$, with $H_{ij} = 1$ if $z_i = j$ and 0 otherwise. One more correction, since we have take the determinant and the inverse of T, so we need necessarily positive definite of T that is not always availlable. We correct this by defining $\hat{T} = T + \sigma^2 Id$ where σ^2 is sufficiently small. This correction will make T automatically positive definite (and of course symmetric). Therefore, our problem turns to

$$\hat{\beta}, \hat{K} \in \arg \min_{\beta \in \mathbb{R}^m, K \in \mathrm{Sym}^+(L)} \log(\det(HKH^T \circ K_X + \sigma^2 Id)) + (Y - F\beta)^t (HKH^T \circ K_X + \sigma^2 Id)^{-1} (Y - F\beta)$$

We will simply write that $T(K) = HKH^T \circ K_X + \sigma^2 Id$, with σ^2 is our job to choose.